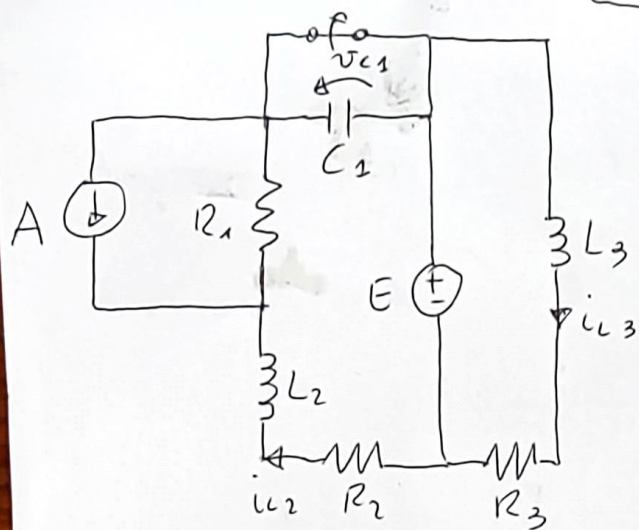


TRANSITORIO



$$C_1 = 3 \text{ mF}, L_2 = 1 \text{ mH}, L_3 = 2 \text{ mH},$$

$$R_1 = 1 \Omega, R_2 = 3 \Omega, R_3 = 1 \Omega,$$

$$A = 2 \text{ A}, E = 3 \text{ V}$$

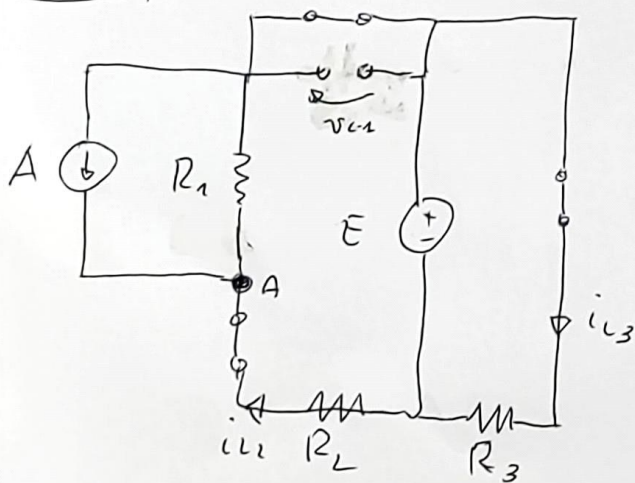
DET:

$$t = 0^- : W_{C1}, W_{L2}, W_{L3}$$

$$t = 0^+ : \frac{dV_{C1}}{dt}, \frac{di_{L2}}{dt}$$

$$t = \infty : P_{R1}, P_E, Q_{C1}$$

$t = \infty$

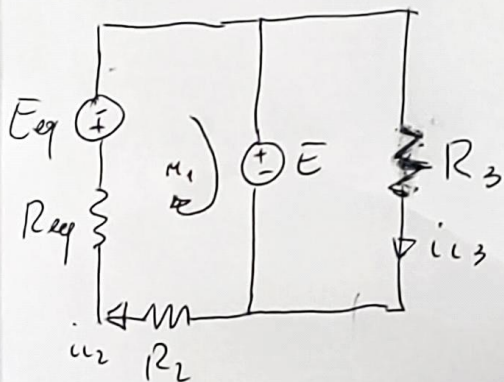


$$i_{L3} = \frac{E}{R_3} = \frac{3}{1} = 3 \text{ A}$$

$$W_{L3} = \frac{1}{2} L_3 i_{L3}^2 = 9 \text{ mJ}$$

$$V_{C1} = 0 \text{ V} \rightarrow W_{C1} = 0 \text{ J}$$

↓



$$E_{eq} = A \cdot R_1 = 2 \text{ V} \quad R_{eq} = R_1 = 1 \Omega$$

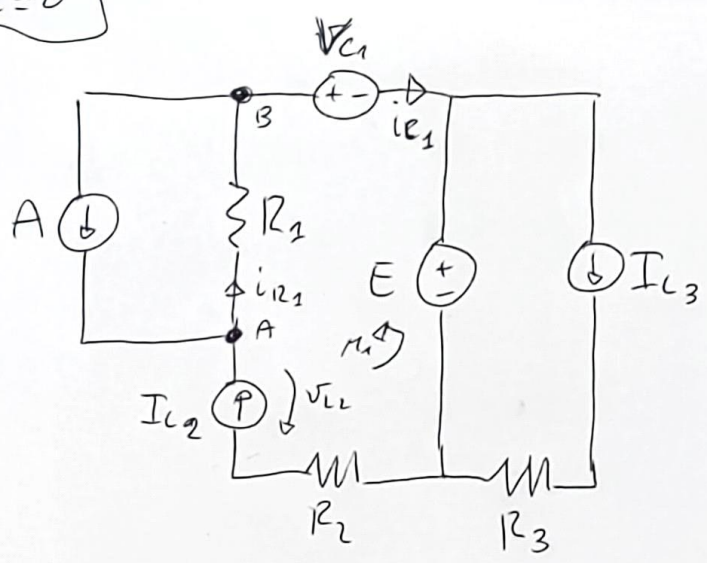
$$\text{LUT } (M_1) : -E_{eq} - E - V_{R2} - V_{R3} = 0$$

$$-E_{eq} - E - R_2 i_{L2} - R_3 i_{L2} = 0$$

$$i_{L2} = \frac{-E_{eq} - E}{R_2 + R_3} = -1.25 \text{ A}$$

$$W_{L2} = \frac{1}{2} L_2 i_{L2}^2 = 0.78 \text{ mJ}$$

$t = 0^+$



$$v_{C1}(t=0^-) = v_{C1}(t=0^+) = V_{C1} = 0 \text{ V}$$

$$i_{L2}(t=0^-) = i_{L2}(t=0^+) = I_{L2} = -1.25 \text{ A}$$

$$i_{L3}(t=0^-) = i_{L3}(t=0^+) = I_{L3} = 3 \text{ A}$$

$$LUC(A) : i_{R1} = A + I_{L2} = 0.75 \text{ A}$$

$$LUC(B) : i_{C1} = i_{R1} - A = -1.25 \text{ A}$$

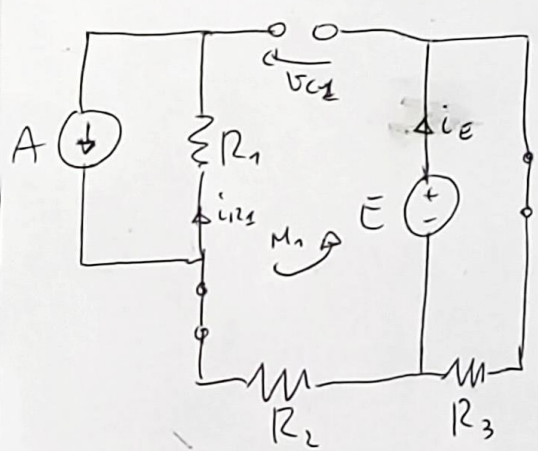
$$\frac{dv_{C1}}{dt} = \frac{i_{C1}}{C_1} = -4.17 \frac{\text{V}}{\text{s}}$$

$$LUT(M_1) : E + v_{C1} + i_{R1}R_1 + v_{L2} + R_2 I_{L2} = 0$$

$$\Rightarrow v_{L2} = -E - i_{R1}R_1 - R_2 I_{L2} = 0 \text{ V}$$

$$\frac{di_{L2}}{dt} = \frac{v_{L2}}{L_2} = 0 \text{ A/s}$$

$t = \infty$



$$i_E = \frac{E}{R_3} = 3 \text{ A}$$

$$P_E = E i_E = 9 \text{ W}$$

$$P_{R1} = R_1 A^2 = 4 \text{ W}$$

$$LUT(M_1) : v_{C1} + v_{R1} + v_{R2} + E = 0$$

$$v_{C1} = -E - R_1 A = -5 \text{ V}$$

$$Q_{C1} = v_{C1} C_1 = -15 \text{ mC}$$

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 [87] M. Khorsany, A. Najati-Chaleli, R. Razzaghi, B. Mohammadi-Ivado, Transactive energy framework for optimal energy management of multi-carrier energy hubs under local electrical, thermal, and cooling market constraints, Int. J. Electr. Power Energy Syst. 129 (2021) 106803.

[88] H. Masrur, M.M. Camil, M.R. Islam, K.M. Murtay, M.H. Lipu, T. Senjyu, An optimized and outage-resilient energy management framework for multi-carrier energy microgrids integrating demand response, IEEE Trans. Ind. Appl. (2022).

REGIME SINUSSIALE

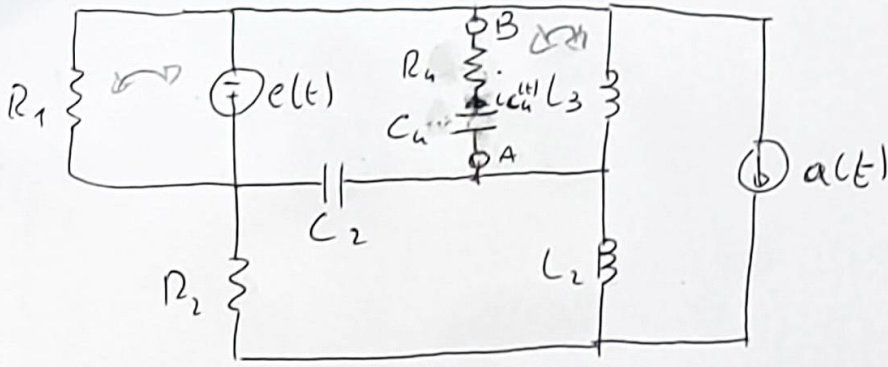
$$R_4 = 4 \Omega, C_4 = 2 \text{ mF}$$

$$R_1 = 2 \Omega, R_2 = 3 \Omega, C_2 = 3 \text{ mF}$$

$$L_2 = 2 \text{ mH}, L_3 = 3 \text{ mH},$$

$$e(t) = 3 \sin(2t - 165^\circ) \text{ V}$$

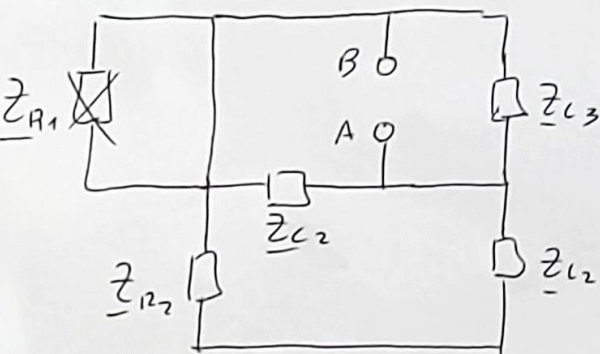
$$u(t) = 6 \sin(2t + 75^\circ) \text{ A}$$



$$\underline{Z}_{R1} = 2 \Omega, \underline{Z}_{R2} = 3 \Omega, \underline{Z}_{C2} = -\frac{j}{\omega C_2} = -166.7 j \Omega, \underline{Z}_{L2} = j\omega L_2 = 4 j \Omega,$$

$$\underline{Z}_{L3} = 6 j \Omega, \underline{E} = \frac{3}{\sqrt{2}} \angle -105^\circ \text{ V}, \underline{A} = \frac{1}{\sqrt{2}} \angle 75^\circ \text{ A}, \underline{Z}_{R4} = 4 \Omega, \underline{Z}_{C4} = -250 j \Omega$$

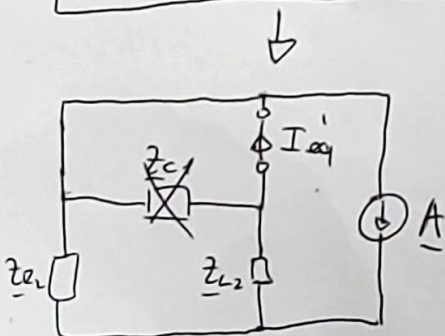
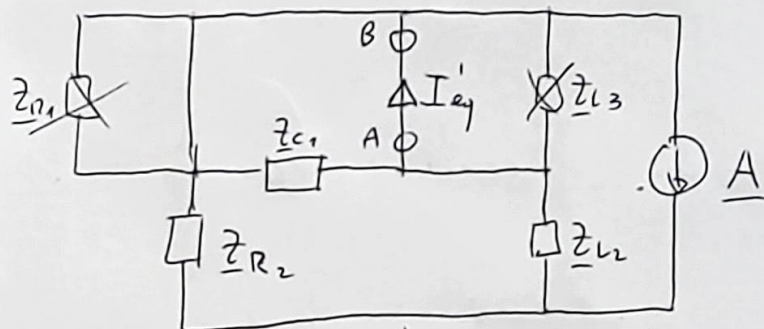
Y_{eq}



$$\underline{Z}_{eq} = \left((\underline{Z}_{R2} + \underline{Z}_{L2}) \parallel \underline{Z}_{C2} \right) \parallel \underline{Z}_{L3}$$

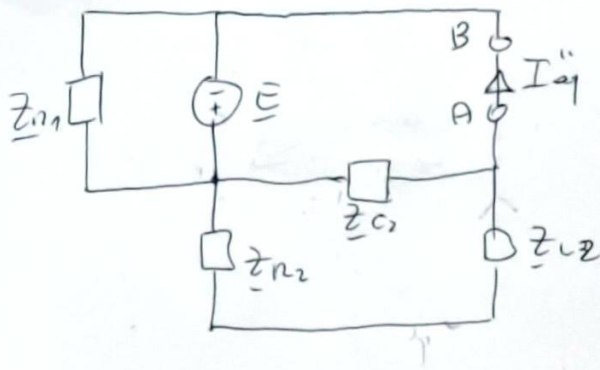
$$\underline{Y}_{eq} = \frac{1}{\underline{Z}_{eq}} = 0.33 - 166.7 j \text{ S}$$

A ≠ 0



$$\underline{I}_{eq} = \underline{A} \frac{\underline{Z}_{R2}}{\underline{Z}_{R1} + \underline{Z}_{L2}} = 0.77 \angle 75^\circ \text{ A}$$

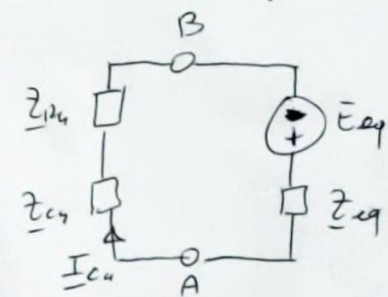
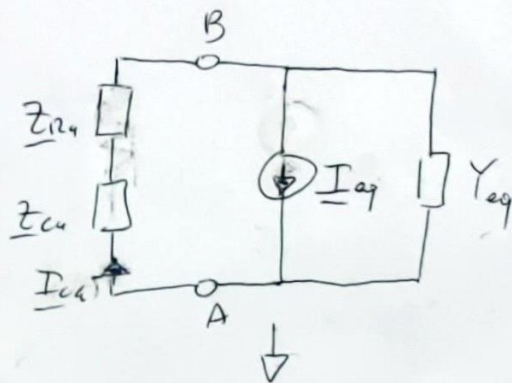
$$\underline{E} \neq 0$$



$$\underline{I}''_{eq} = \frac{\underline{E}}{\underline{z}_{L2}} + \frac{\underline{E}}{\underline{z}_{n1} + \underline{z}_{L2}} = \underline{E} \cdot \dots$$

$$= 0.71 \angle 106^\circ$$

$$\underline{I}_{eq} = \underline{I}_{eq}' + \underline{I}_{eq}'' = 1.36 \angle 90^\circ \text{ A}$$



$$\underline{I}_{cu} = + \frac{\underline{E}_{eq}}{\underline{z}_{eq} + \underline{z}_{n1} + \underline{z}_{cu}} = + \frac{\frac{\underline{I}_{eq}}{\underline{Y}_{eq}}}{\frac{1}{\underline{Y}_{eq}} + \underline{z}_{n1} + \underline{z}_{cu}} =$$

$$= +3.27 \cdot 10^{-5} \angle -90^\circ \text{ A}$$

$$i_{cu}(t) = 4.62 \cdot 10^{-5} \sin(2t + 90^\circ) \text{ A}$$

$$\underline{N} = \underline{z}_u |\underline{I}_{cu}|^2 = (R_u + jX_{cu}) |\underline{I}_{cu}|^2 = 4.24 \cdot 10^{-5} - 2.67 \cdot 10^{-7} j \text{ VA}$$